Charles Krebs

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EDUCATION

Tufts University, Medford, MA — Expected May 2027
Bachelor of Science in Mechanical Engineering, Minor in Music Engineering
GPA: 3.30 | Dean's List (Fall 2023, Spring 2024, Fall 2024, Fall 2025, Spring 2026)
Relevant Coursework: Acoustics; Acoustic & Electronic Instrument Design; Intro to Music Engineering; Electronics & Controls; Thermal Fluid Systems; Engineering Design; Materials & Manufacturing; Mechanics; Music Theory

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, Onshape, SolidWorks Simulation & Motion, COMSOL, SketchUp
Acoustics & Audio: CATT-Acoustic, Max/MSP, Reaper, Room EQ Wizard, measurement microphones, impulse response measurement, auralization, FFT/PSD analysis
Programming & Analysis: Python, CircuitPython, MATLAB, C++, LabVIEW
Electronics: KiCad, PCB design, soldering, oscilloscope, multimeter, op-amp circuits, Arduino / Raspberry Pi, lidar sensor integration
Fabrication: Laser cutter, 3D printing, lathe, drill press, power saws, waterjet, injection molding
Mechanical Testing: Instron tensile tester, strain gauge, flow sensor, thermocouple/thermistor, load cell

RELEVANT PROJECTS

Acoustical Analysis of a Multipurpose Hall, Architectural Acoustics — Spring 2026
- Measured impulse responses in a 27,000 ft³ campus hall and built a Sabine model that predicted reverberation time within 10% in every octave band
- Modeled and auralized two redesigns in CATT-Acoustic, showing 3" fiberglass panels would cut low-frequency reverberation time from 1.62 s to 1.03 s for clearer speech

Cordless Drill Noise Mitigation, ME 126 — Fall 2025
- Designed and tested foam, HDPE, and combined noise barriers, analyzing recordings at 0.75 m and 1.0 m with a custom MATLAB spectral analysis script
- Reduced drill noise by up to 9.0 dBA with the combined barrier, more than double the best single material

Banjo Design & Head-Tension Acoustics, Instrument Design — Spring 2026
- Designed and hand-built a playable five-string banjo from raw lumber, including a laminated rim, carved neck, and custom tension-hook mounting system
- Verified that head fundamental frequency scales with √tension across 6 tension levels ($\sigma < 8$ Hz) against a COMSOL FEA model, attributing the slope difference to air loading

Remote-Controlled Lifting & Weighing Robot, Electronics & Controls — Fall 2025
- Built a three-motor robot that drives, lifts a crate with a belt-driven arm, and sorts it into four weight classes using a load cell and op-amp amplifier
- Wrote a Flask web interface on a Raspberry Pi for browser control over Wi-Fi, with a KB2040 microcontroller handling analog sensing over GPIO

PCB Design: Power Supply & H-Bridge Motor Driver, Electronics & Controls — Fall 2025
- Designed and fabricated a breadboard power supply PCB delivering 12 V, 5 V, and 3.3 V simultaneously from a 12 V adapter
- Designed a MOSFET H-bridge with BJT gate drivers for 3.3 V logic, rated for 12 V at 5 A; validated bidirectional motor control on a breadboard before layout

Reciprocating Saw Design Evaluation, Engineering Design II — Fall 2025
- Analyzed a reciprocating saw's driving pin, spiral bevel gears, and bearings for static, fatigue, contact, and bearing-life failure using Shigley's and AGMA methods
- Identified the gear train as the limiting component; FEA matched hand-calculated pin stress within 17%.

Pin & Pull Assistive Device, Engineering Design I Spring 2025

- Co-designed a two-part assistive tool that lets a user with cerebral palsy place pins on her pin board independently, translating 8 user needs into engineering requirements
- Ran think-aloud usability tests with the user, who went from needing heavy assistance to completing the task largely independently by the fourth trial

Product Teardown: Oscillating Multi-Tool, Engineering Design I Spring 2025

- Disassembled a corded multi-tool, documenting 15 part types with their materials and manufacturing processes, plus an 8-step disassembly procedure
- Modeled 5 major subassemblies as a full SolidWorks assembly with an exploded-view animation

Gait Inverse Dynamics Model, Mechanics II Spring 2025

- Built a three-segment inverse-dynamics leg model computing ankle, knee, and hip forces and torques from motion-capture and force-plate data
- Found peak ankle push-off torque was roughly 3× peak hip extensor torque across the stance phase

EXPERIENCE

Idea Lab, Tufts University, Medford, MA — *Student Researcher* Sep 2025 – Present

- Lead sound integration for a wearable lidar-based navigation device that helps visually impaired users perceive their surroundings through audio
- Design an audio feedback system that conveys obstacle distance and direction through varying pitch, timbre, and beep rate
- Write Python software that processes stacked lidar scan planes into a map of the user's surroundings and turns it into real-time audio cues
- Contribute to hardware design and fabrication of the wearable prototype, including CAD, 3D-printed enclosure, sensor mounting, wiring

Tufts Center for Engineering Education & Outreach, Medford, MA — *Teaching Fellow* Jan 2026 – Aug 2026

- Developed 5 weeks of music engineering curriculum and hands-on projects covering transducers (speakers and microphones), electronic and acoustic instruments, architectural acoustics, and human hearing
- Taught a 2-week program for ~35 high school students and a 1-week program for ~60 middle school students
- Explained acoustics and audio concepts to students with no technical background

Lost Nation R&D, Albany, VT — *Design & Fabrication Intern* Jun 2022 – Sep 2022

- Manufactured biathlon rifle stocks, straps, and mounting hardware for commercial sale to national teams and individual athletes worldwide
- Ran small-batch production using a lathe, laser cutter, and power tools, then assembled finished rifles and prepared them for shipment

Manufacturing Solutions Inc., Morrisville, VT — *Assembler* Jun 2024 – Sep 2024

- Assembled Concept 2 Ski-Ergs to specification
- Gained hands-on experience in design for manufacturability and assembly by seeing how part design affected assembly time and precision

NEK Biosciences, Barton, VT — *Research Assistant* Jun 2024 – Sep 2024

- Researched how common pressure ulcers are in dairy cattle and their economic impact, to assess the market for a new product
- Presented findings on market size and potential applications

Somerville Arts Council, Somerville, MA — *Volunteer* Jul 2025 – Sep 2025

- Set up and took down stage and audio equipment for community music events; staffed the attendee information table

Hosmer Point Camp, Craftsbury, VT — *Camp Counselor* May 2023 – Aug 2023

- Led daily outdoor activities and supervised groups of 8–14 campers per session

Property Care & Landscaping, Craftsbury, VT — *Owner* May 2018 – Aug 2022

- Founded and ran a seasonal property-care business for 4 clients, handling scheduling, billing, and client communication

ACTIVITIES & ATHLETICS

Tufts Varsity Cross Country & Track and Field — *NCAA Division III Athlete* Sep 2023 – May 2026

- Trained 15–20 hours per week while carrying a full mechanical engineering course load

Applejam, Tufts University — *Member* Jan 2025 – Present

- Book local bands and provide live sound setup and support for 5 – 7 shows per semester, including equipment setup and mic placement

WMFO Radio, Tufts University — *DJ* Jan 2025 – Present

- Host a weekly freeform radio show, running a live broadcast studio with microphones, turntables, CD players, and a mixing board